

crossed product by
modular group $\sigma_s = \Delta_{\Psi}^{-is} \cdot \Delta_{\Psi}^{is}$

$\text{tr}(\hat{A}) = \langle \hat{\Psi} | \hat{A} \rho^{-1} | \hat{\Psi} \rangle$
density operator $\rho_{\hat{\Phi}}$ exists

$\mathcal{M}$
type III₁, no trace

$\widehat{\mathcal{M}} = \{e^{iK\hat{p}} A e^{-iK\hat{p}}, e^{-i\hat{q}s}\}''$
type II_∞ (or II₁ if $\hat{q} \geq 0$): *has* a trace

clock $L^2(\mathbb{R}_{\hat{q}})$
 $[\hat{q}, \hat{p}] = i$

entropy $S_{\widehat{\mathcal{M}}} = -S_{\mathcal{M}}(\Phi \| \Psi) + \text{const}$
= generalized en-
tropy S_{gen} (gravity)